AIP-Suite User Guide

AIP-Scope and three desktop analysis modules

29 October 2026

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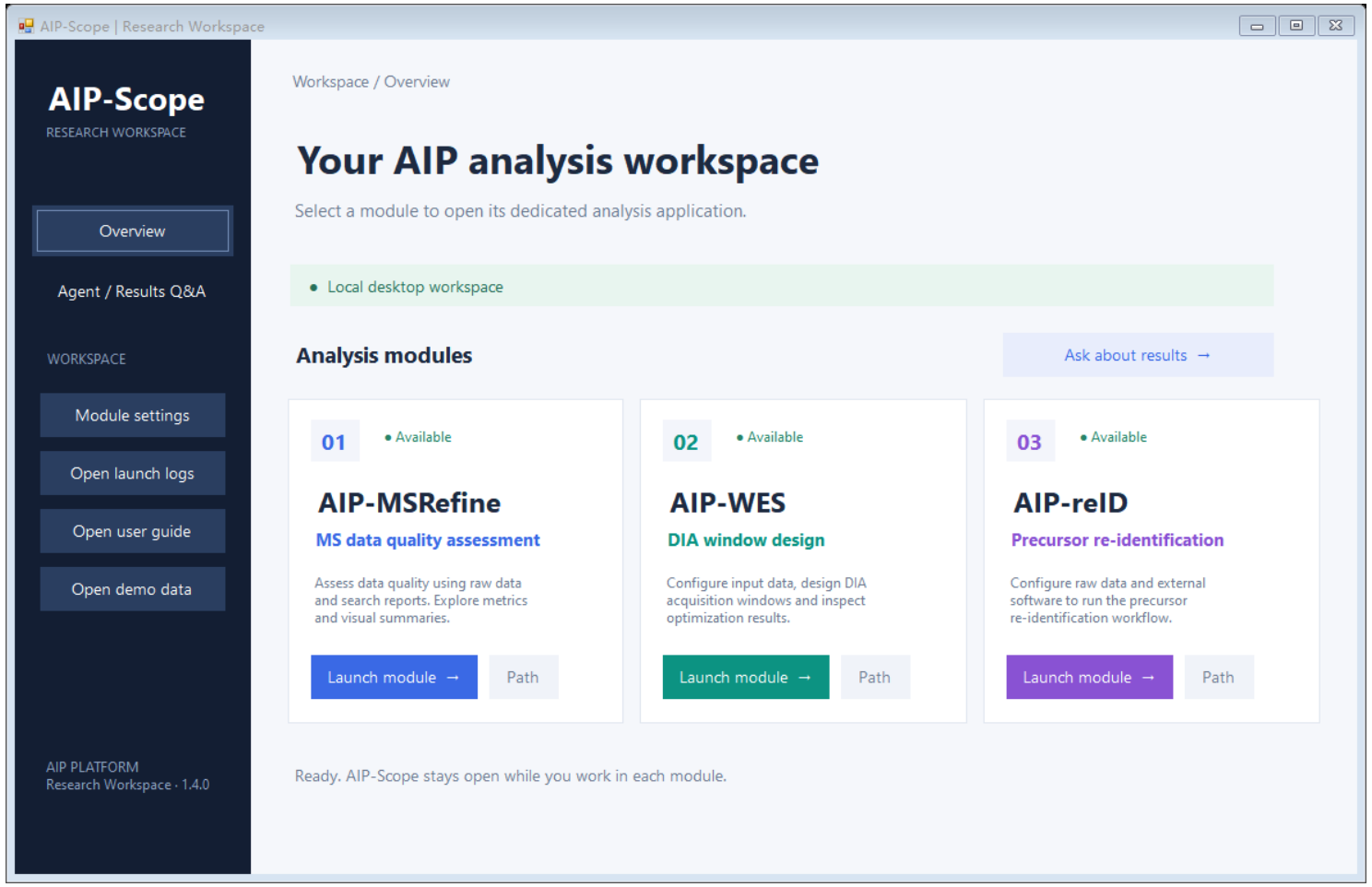
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1. AIP-Scope main interface



1. Open AIP-Scope.exe in the suite folder. Launch module opens the selected application in its own window.

2. Use Open user guide for this manual and Open demo data for the separate demonstration data folder.

3. The suite resolves its bundled module folders. Use Path or Module settings when selecting another installation.

Available means that the EXE exists. Launched describes an active application process, not analysis completion.

2. Installation and module paths

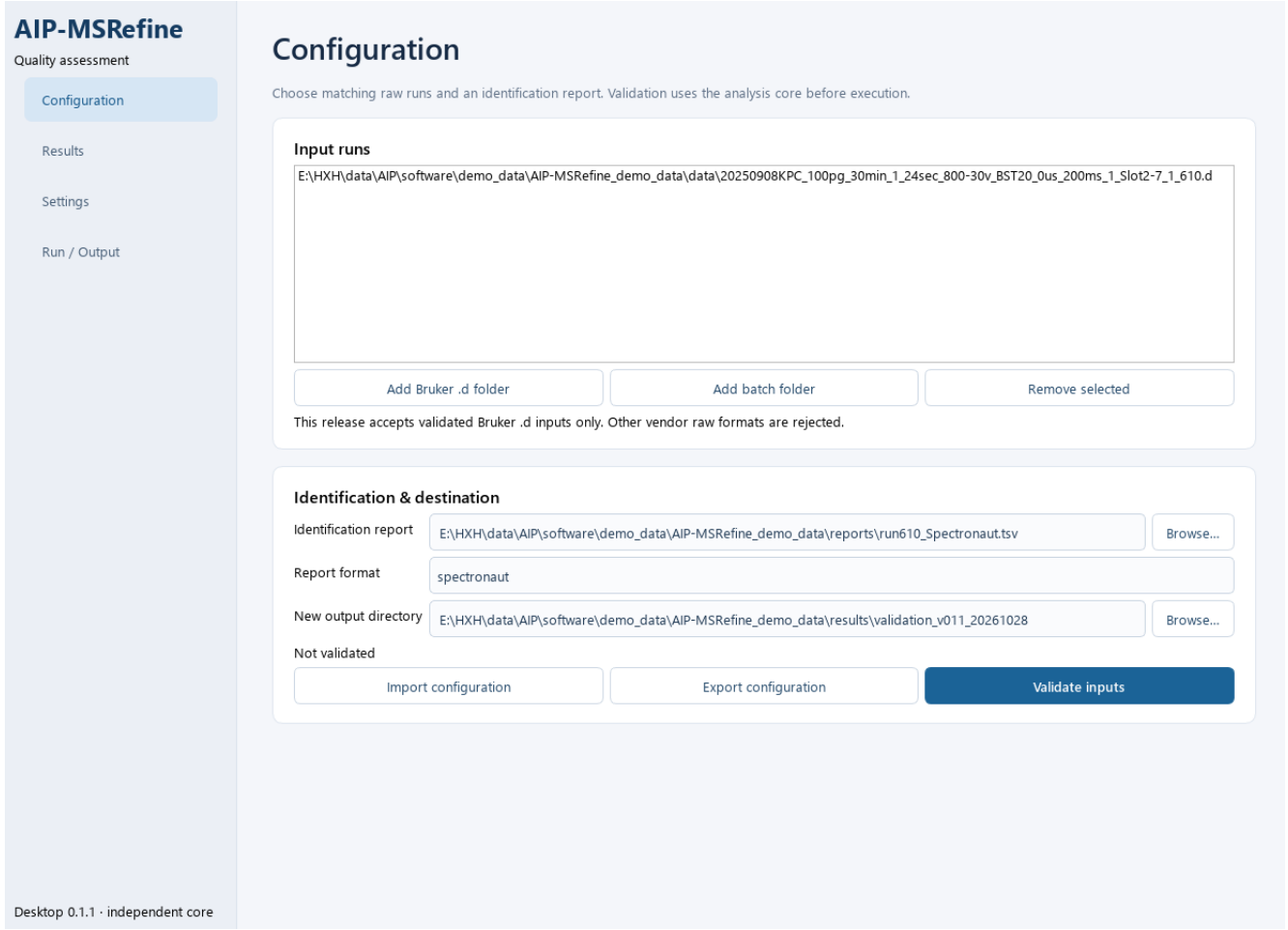
Extract the complete AIP-Suite program ZIP. AIP-Scope.exe is at the root. The modules/AIP-MSRefine, modules/AIP-WES and modules/AIP-reID folders contain the complete applications and their Python runtimes.

Keep each GUI, Core executable and runtime directory together. WES needs both _internal and _gui_internal. AIP-reID additionally requires a separately installed, complete DIA-NN 2.3.2 distribution.

Demo data are distributed separately from the program ZIP. Open demo data uses demo_data inside the suite when present, otherwise the sibling demo_data folder. Keep each demo folder complete when moving it.

AIP-Scope requires .NET Framework 4.5 or later. Windows 10/11 supports model-profile import. Closing the launcher leaves analyses running. Each run needs fresh output and sufficient space; reID requires a short local output path.

3. AIP-MSRefine input data



1. Use Open_Demo.ps1 in AIP-MSRefine_demo_data to open the demo, or Add Bruker .d folder for your own input.
2. Choose the matching Identification report and set Report format to spectronaut or diann.
3. For a new run, select an empty New output directory, then Validate inputs. Correct any reported errors.

Normal startup uses empty inputs and default parameters. Import configuration or Open_Demo.ps1 fills them explicitly. Demo: run610 Spectronaut. Keep the complete .d acquisition.

4. AIP-MSRefine report preparation

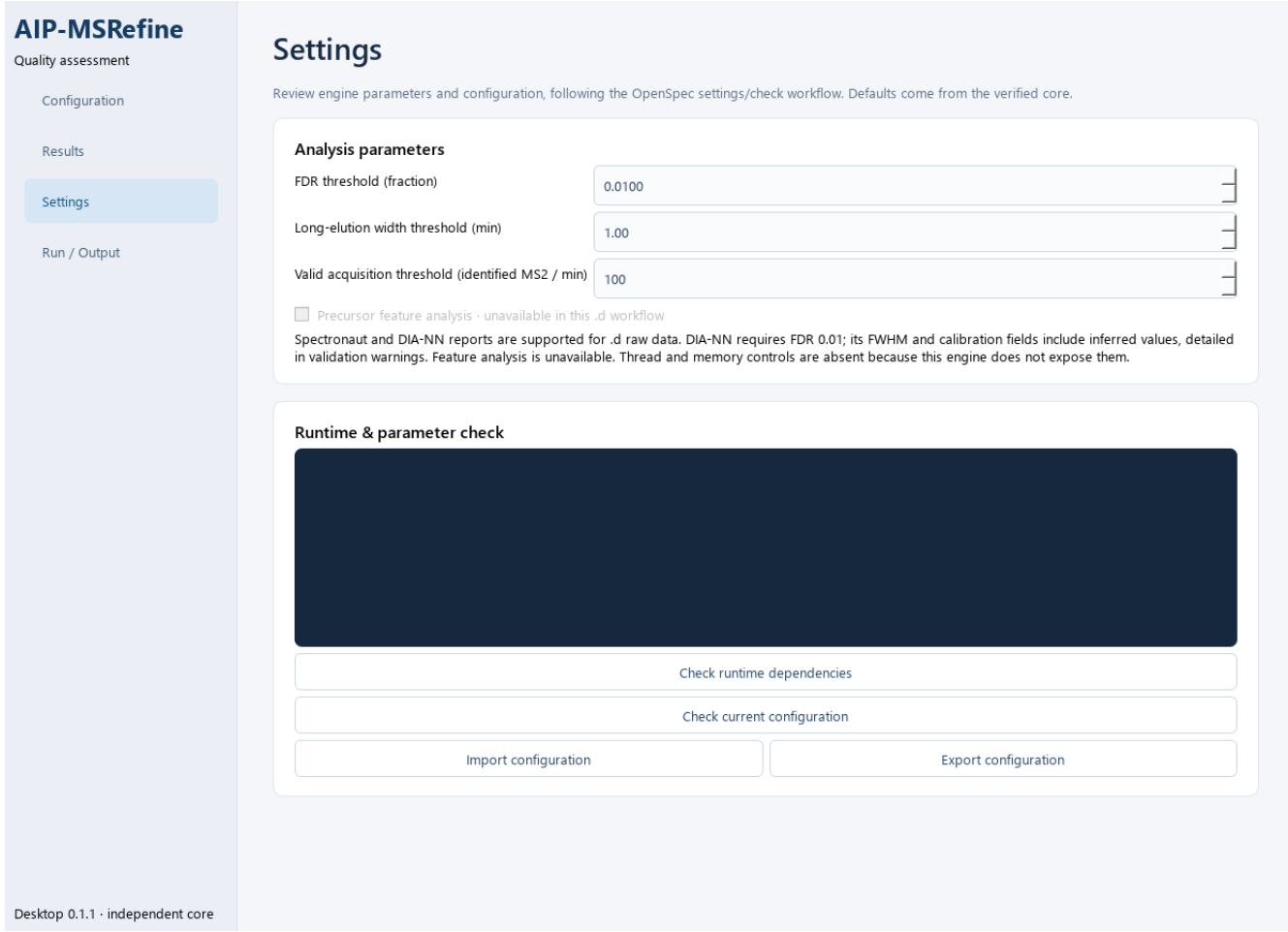
Use a tab-separated TSV or TXT report from the same acquisition. Minimal quantification exports and Parquet files are insufficient. The validator lists missing columns before execution.

For Spectronaut, export the required precursor identity, RT, quantity, FDR, charge and fragment fields. See page 28 for the full column list, including EG.IsDecoy and FG.RawMassAccuracy (PPM).

For DIA-NN, retain Run or File.Name, Precursor.Mz and the complete RT, quantity and FDR fields listed on page 29. The preserved parser requires a precursor Q-value threshold of 0.01.

Match report run names to the selected raw data and verify sample identity. Report RT uses minutes. Begin with one run. Multi-run batch execution has not completed a full validation in this release.

5. AIP-MSRefine analysis settings



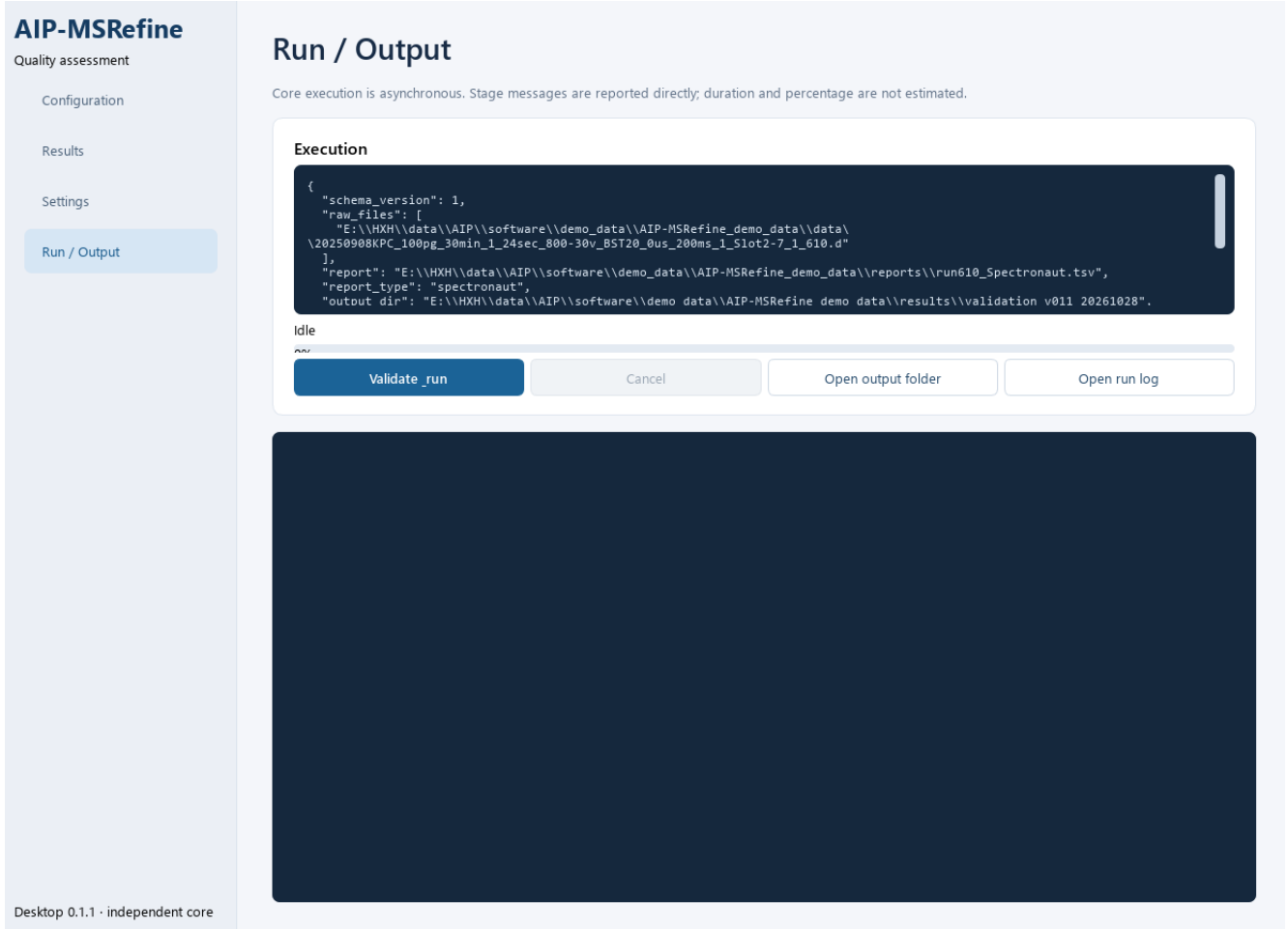
1. Open Settings. Review FDR threshold and the two acquisition or elution thresholds.

2. Use Check runtime dependencies and Check current configuration before running.

3. Export configuration to save the input paths and parameters for reproducibility.

Defaults: FDR 0.01, long-elution width 1 min and valid acquisition threshold 100 identified MS2/min.

6. AIP-MSRefine execution



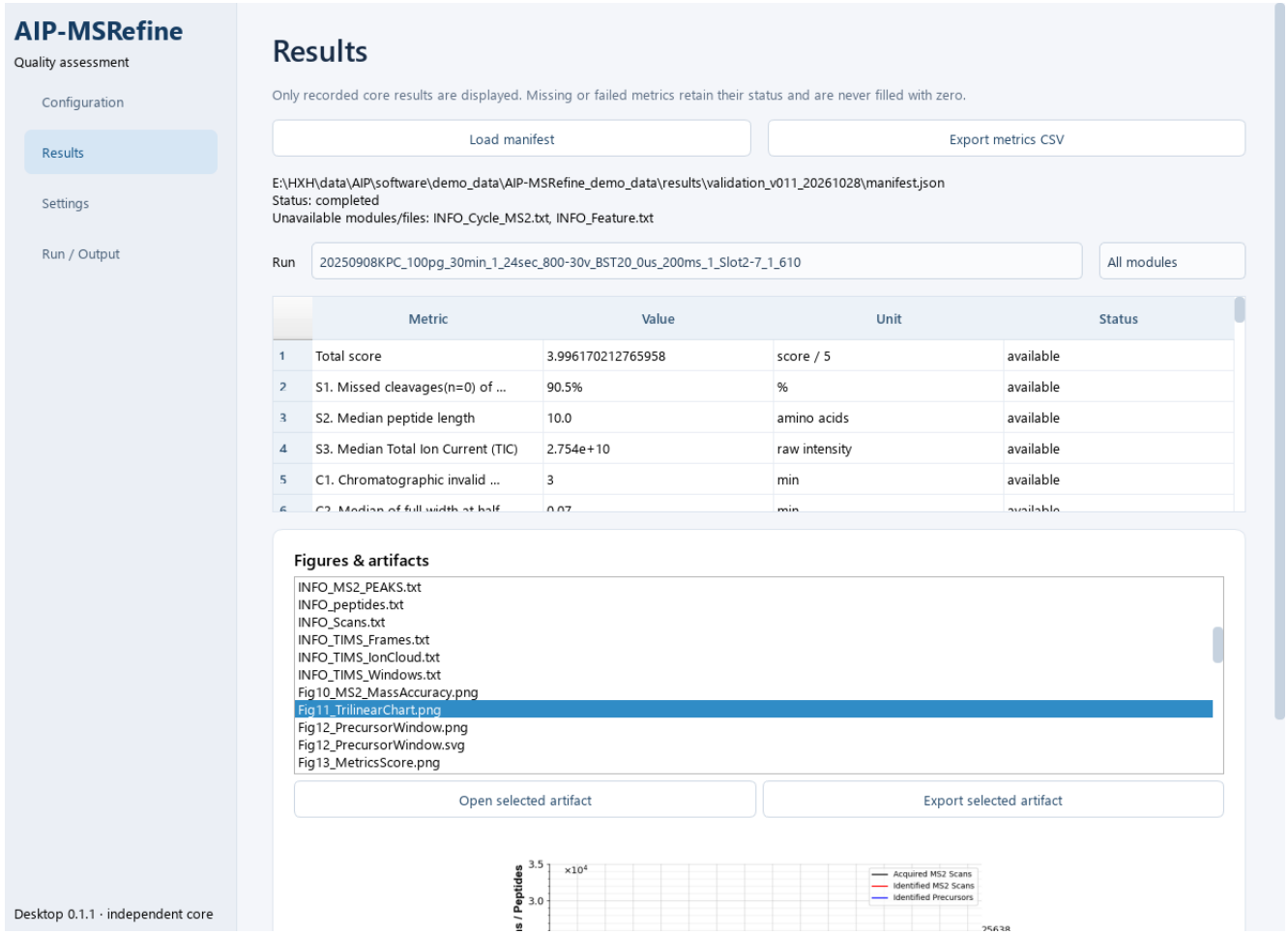
1. Open Run / Output and check the parameter summary. Click Validate & run when ready to start a new analysis.

2. Follow the stage label and console messages. Extraction progress covers that stage only.

3. Use Open run log for details. Cancel stops the active process tree and retains partial files.

Idle GUI capture. The supplied demo ran through Core 0.1.1 and its manifest was loaded afterward; this image is not a GUI execution receipt.

7. AIP-MSRefine result overview



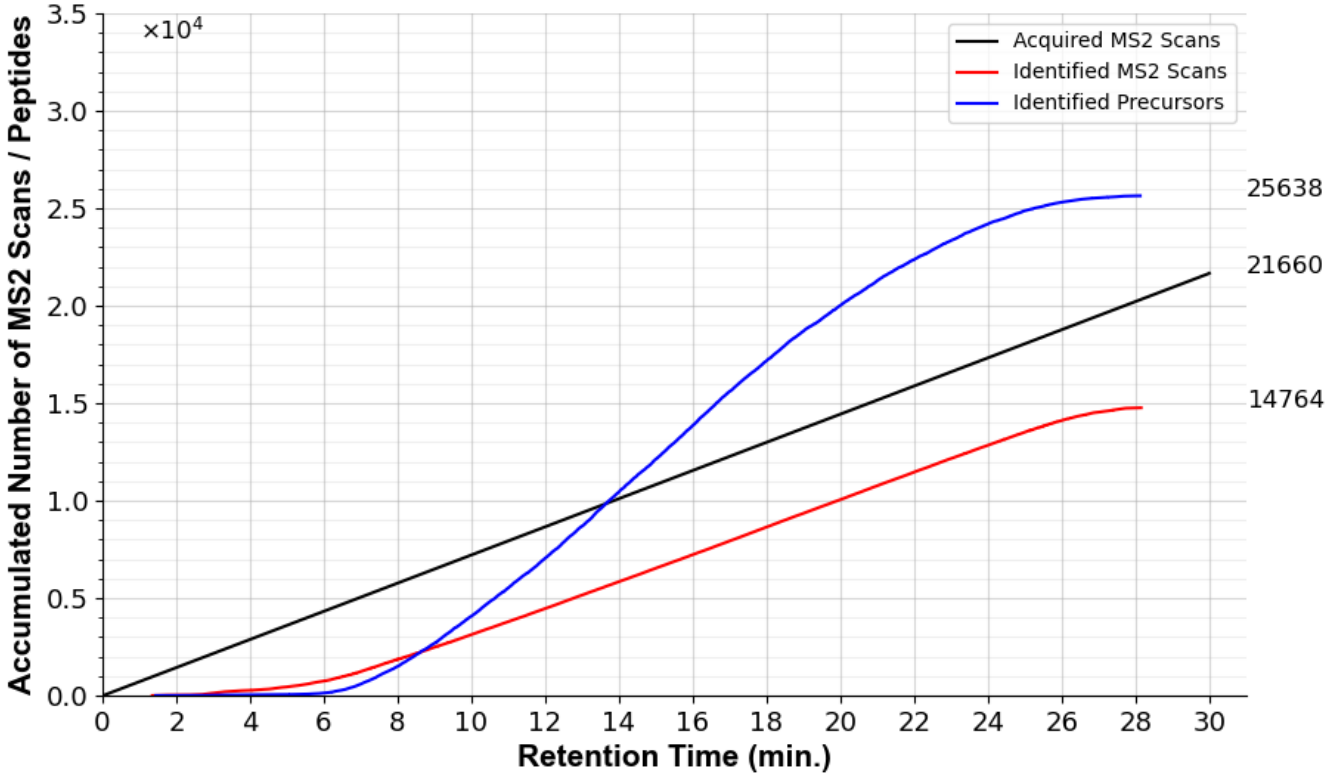
1. Open Results and use Load manifest to select manifest.json from a completed output folder.

2. Select the Run and an analysis category. Read each Metric together with its Unit and Status.

3. Hover over a metric for its definition. Use Export metrics CSV to save the displayed metrics.

Actual Core 0.1.1 run610 result loaded into the GUI. INFO_Cycle_MS2 and INFO_Feature remain unavailable; missing values are not zero-filled.

8. AIP-MSRefine plots and artifacts



1. In Results, select an item in Figures & artifacts. Scroll down to inspect its preview.

2. Use Open selected artifact for the full view, or Export selected artifact to save a copy.

3. In the three-line plot, distinguish acquired MS2 window spectra, identified MS2 spectra and unique precursors.

Actual 0.1.1 run610 plot: 21,660 acquired window spectra, 14,764 identified MS2 spectra and 25,638 unique precursors.

9. AIP-MSRefine saved results

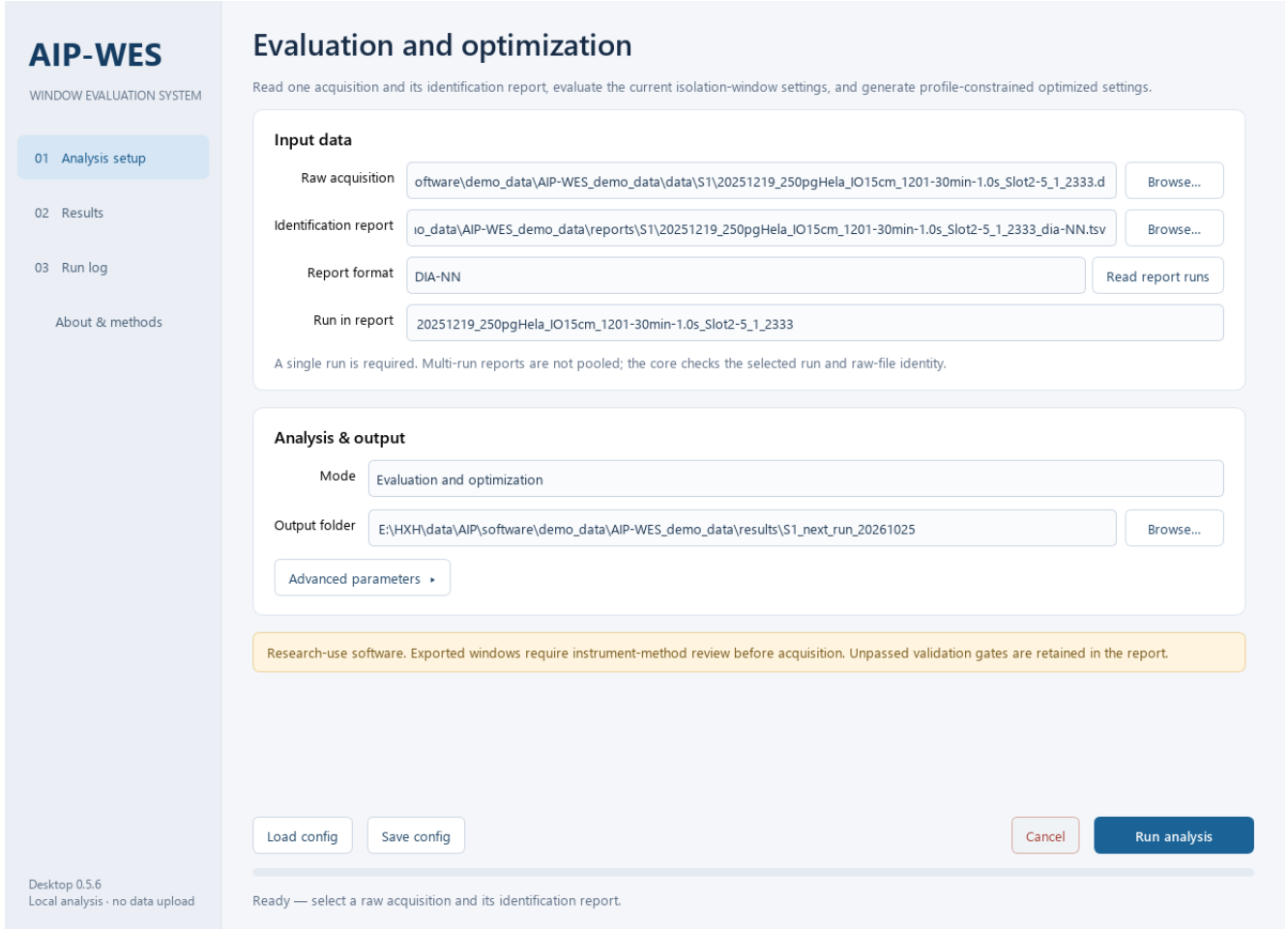
manifest.json links to the complete MSCohort_<run> folder with native tables, figures and the HTML report. GUI table previews show up to 500 rows. Preview this demo at results/validation_v011_20261028/manifest.json.

The integrated Core 0.1.1 completed a fresh run610 analysis. An independent audit passed for 25,638 accepted precursors, 127 metric rows and 59 artifacts. GUI checks loaded this manifest; they did not execute a new GUI run.

Extraction checks matched 1,445 MS1 spectra and 21,660 DIA window spectra across 7,220 DIA frames. No frames were skipped; RT differences from native metadata were zero. MS2 spectrum counts are distinct from precursor counts.

Keep configuration.json and run.log with all results. Source NaNs and unavailable metrics retain their status. AIP-SOP scores are QC summaries. DIA-NN FWHM is estimated; a full DIA-NN/raw-data demo is not validated here.

10. AIP-WES input data



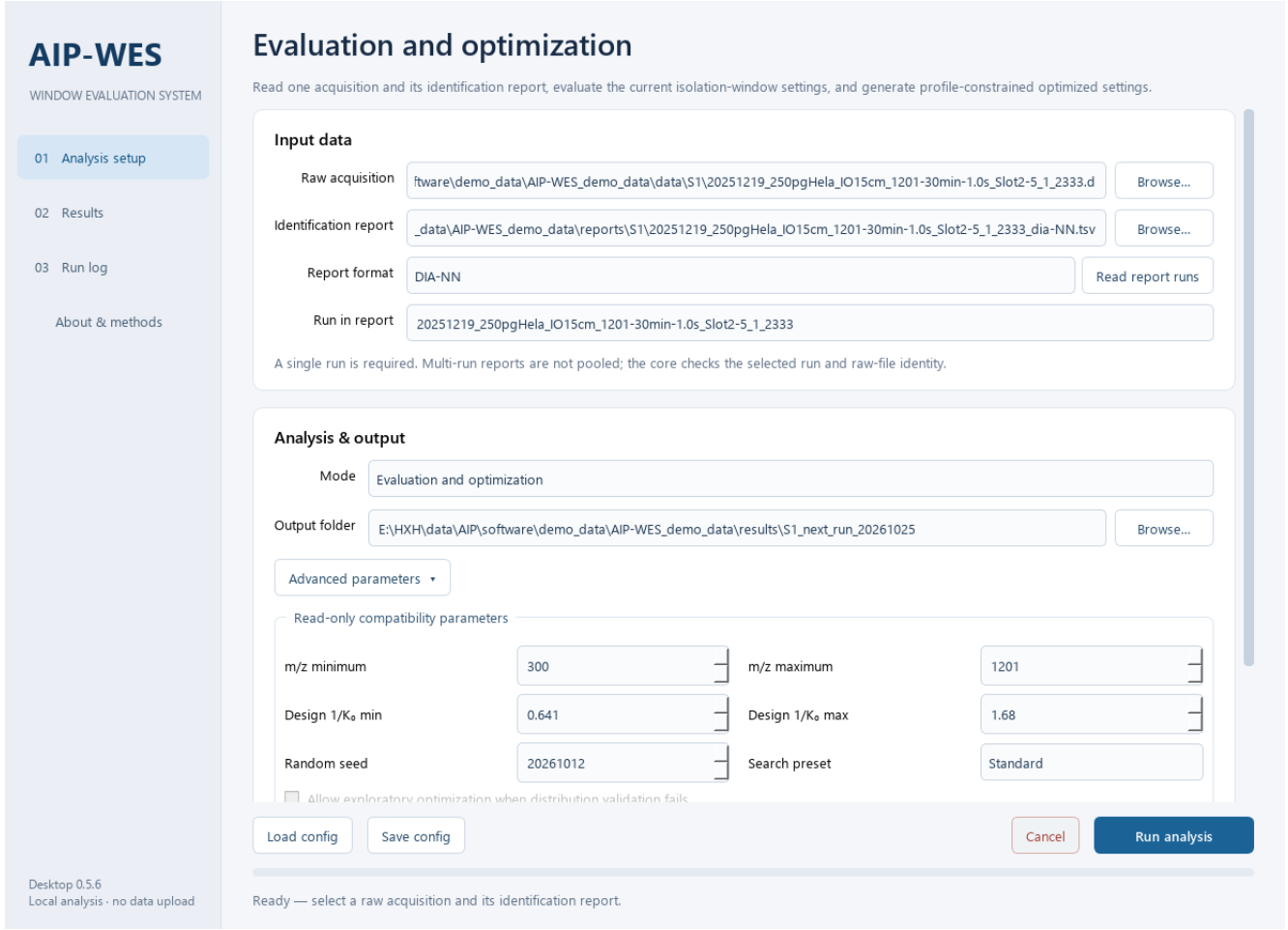
1. Normal startup clears prior inputs. Use Load config, or select the .d acquisition and matching report.

2. Use Read report runs, select the exact run, then choose the evaluation mode.

3. Choose a fresh Output folder. Evaluation and optimization uses the packaged reference profile.

Open_Demo.ps1 -Sample S1 fills demo settings explicitly. Rerun the helper after relocation. Fixed repeated diaPASEF windows and measured inverse mobility are required.

11. AIP-WES parameters and mode



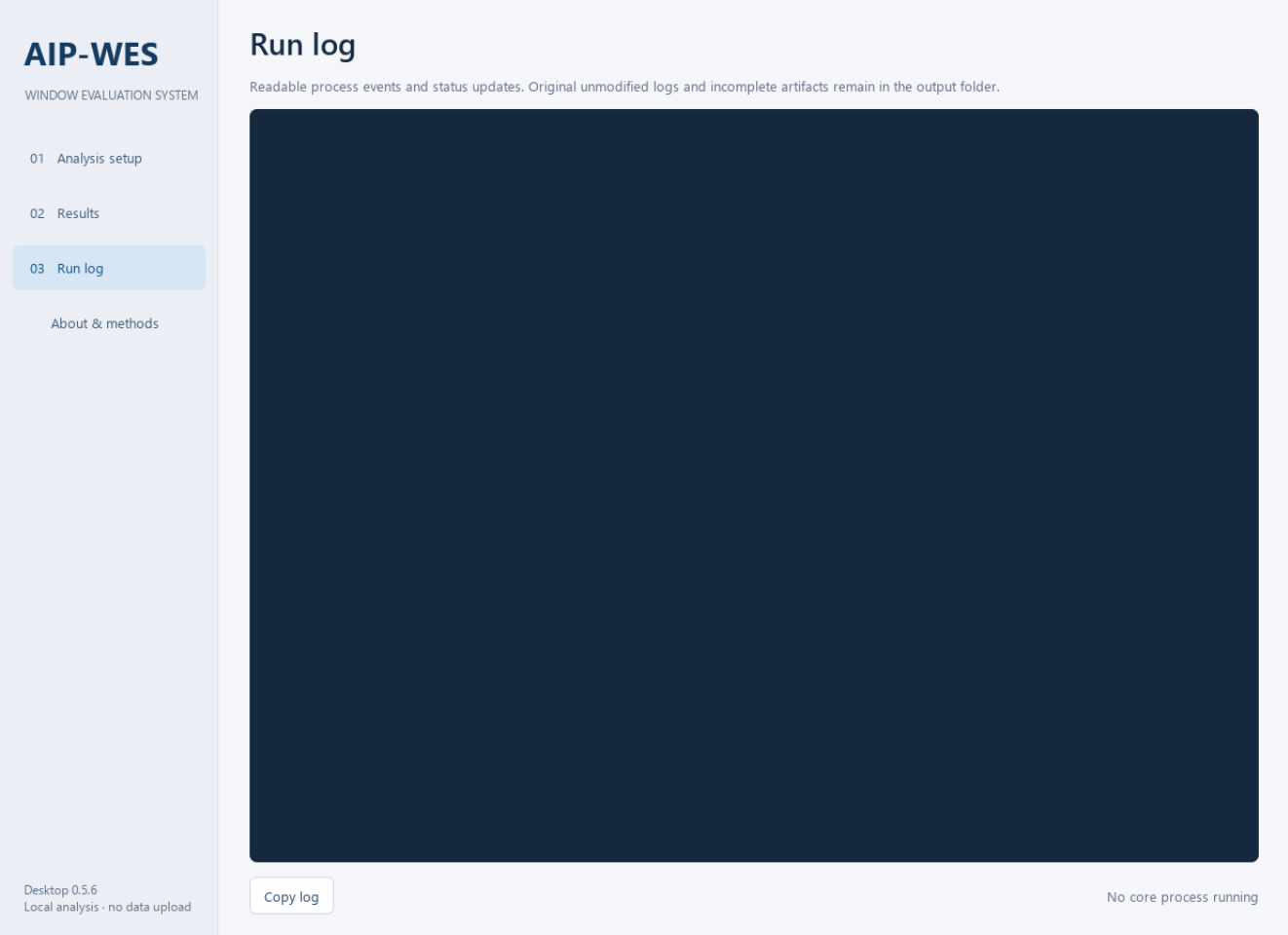
1. Expand Advanced parameters to inspect read-only compatibility values. TIMS ramps and stages are hidden in this version.

2. Search preset, random seed and the exploratory override do not control the current deterministic profile branch.

3. Imported values must still be legal. Design IM bounds must fit the raw calibration range; they do not set the profile output.

Fitting domain: m/z 300–1201 and 1/K0 0.60–1.70. The packaged 5-group/3-stage profile governs candidate topology.

12. AIP-WES current execution



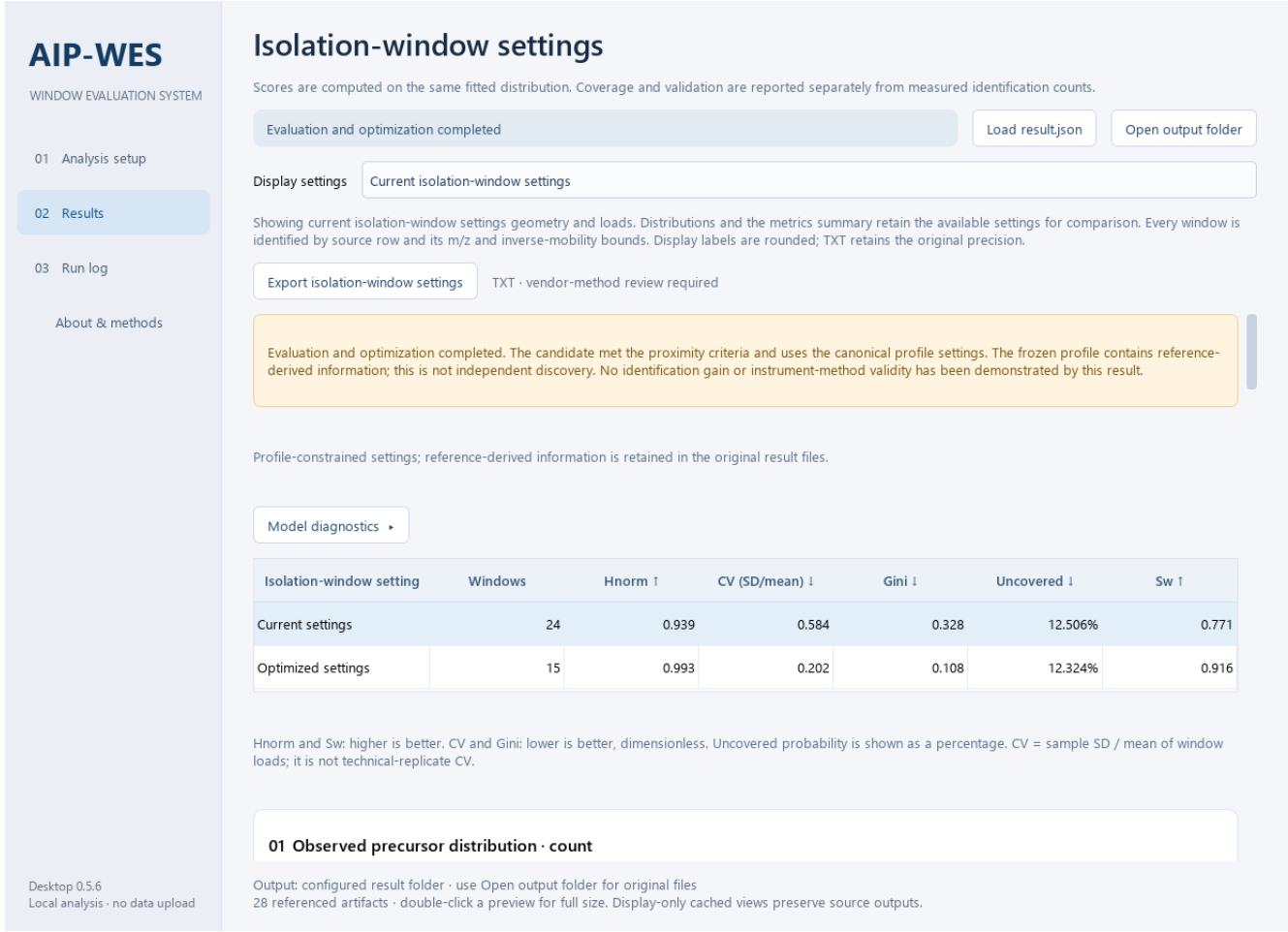
1. In Evaluation and optimization mode, Run analysis fits this run anew, scores current windows and checks S3 proximity.

2. When needed, it solves reference quantiles, projects IM bounds to raw support, then checks merge criteria and final geometry.

3. Follow Run log. Cancel retains partial files. A usable fit may finish with scientific warnings that still require review.

Evaluation only fits and scores the input; it does not calibrate to S3. Image: idle Run log. No independent discovery or measured ID gain is established.

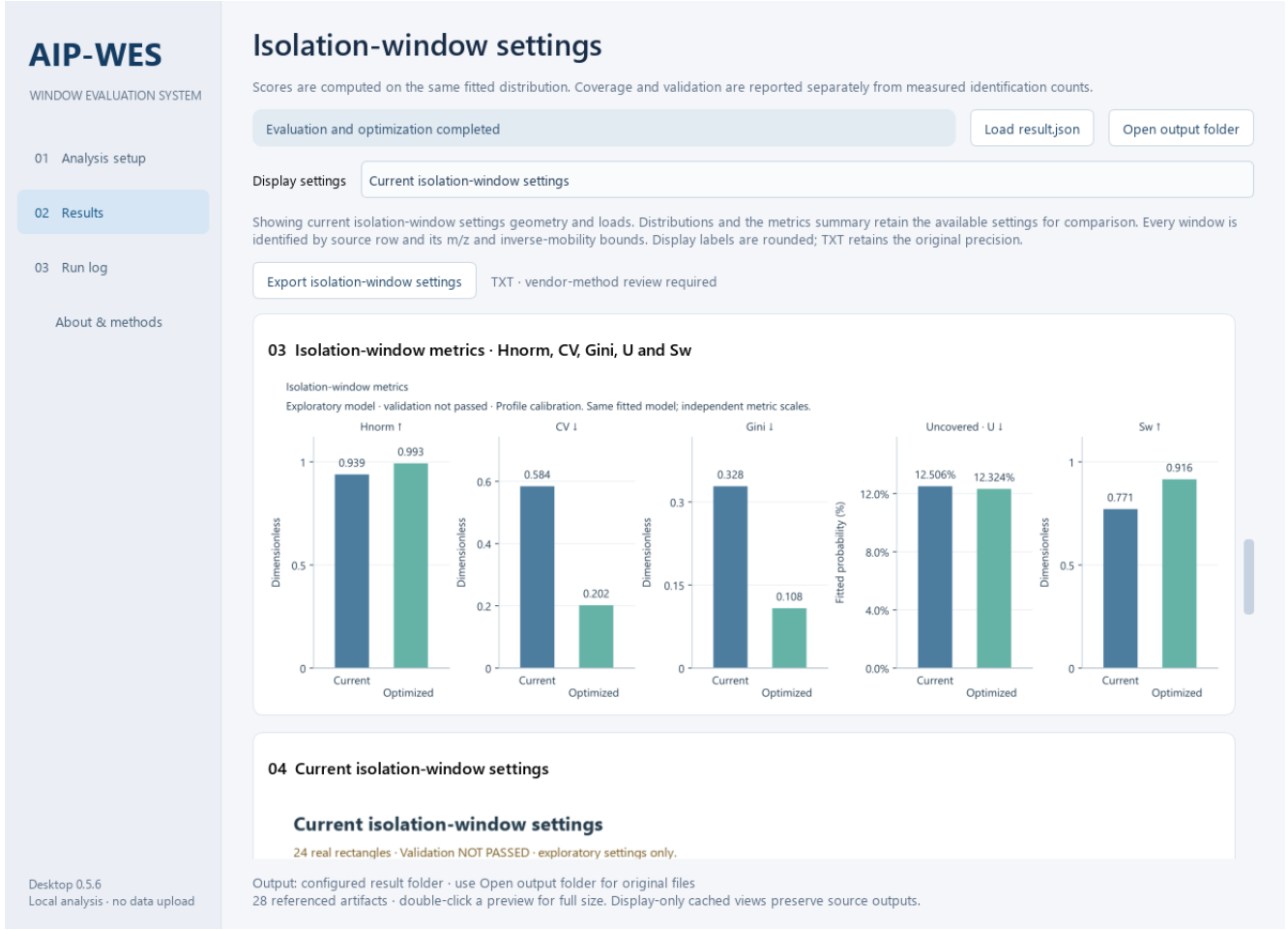
13. AIP-WES result overview



1. Open Results or Load result.json. Check the completion message and expand Model diagnostics for the full validation state.
2. Choose Current isolation-window settings to inspect the input geometry and its measured or modeled loads.
3. Compare the candidate with this same run. Failed scientific checks remain recorded even when numerical execution completes.

Example: saved S2 result. Model diagnostics retains detailed fit warnings. Numerical completion does not clear the recorded scientific limits.

14. AIP-WES optimized settings



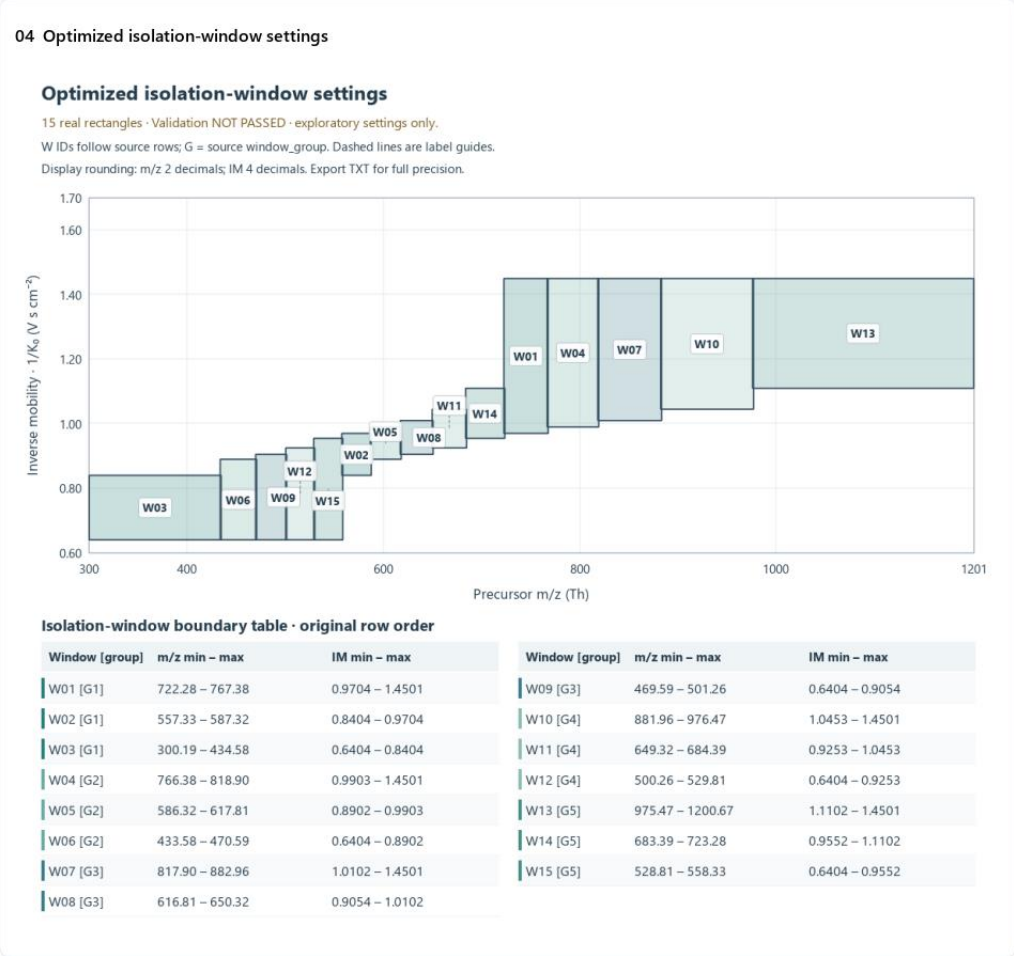
1. Choose Optimized isolation-window settings to inspect the candidate saved by the current reference-calibrated workflow.

2. Compare Hnorm, sample CV, Gini, uncovered probability and Sw. Metrics display 3 decimals; JSON/TSV retain full precision.

3. Double-click a preview for a full-size figure. Changing Display settings does not rerun fitting or candidate generation.

Example: S2. Observed density means precursor counts; fitted density means probability. Retain-current advice does not change the exported windows.

15. AIP-WES window export



1. KPC run858 has 15 current and 15 candidate windows. IM edges change slightly; m/z edges do not.

2. Export isolation-window settings saves the six-column physical table at its original precision.

3. Retain-current advice does not rewrite the export. Fit validation still failed; review the geometry in vendor software before acquisition.

KPC15: Sw decreases 0.737096171 → 0.737008137; max IM change ≈ 0.0002713. Columns: window_group, window_index, mz_min, mz_max, im_min, im_max.

16. AIP-WES score interpretation

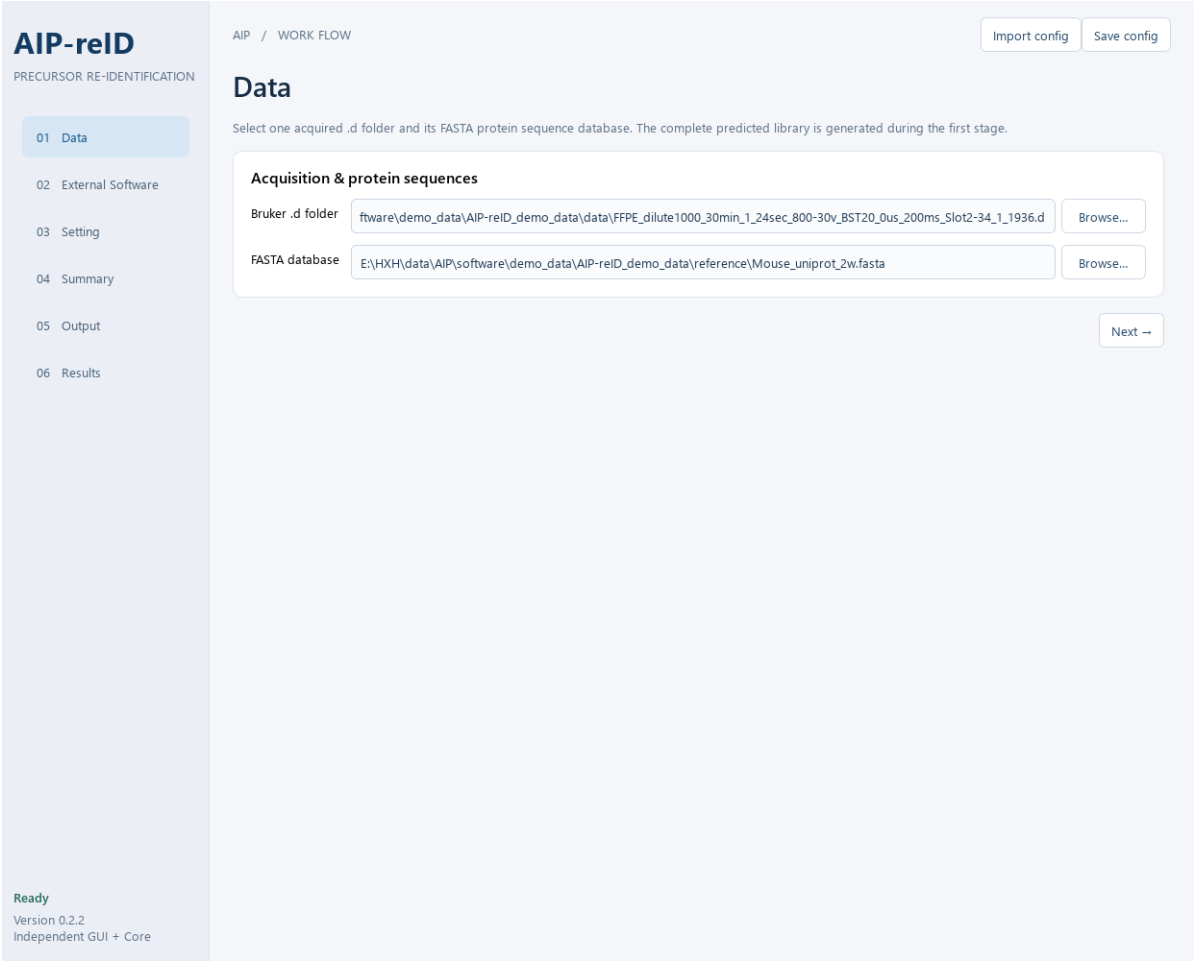
Hnorm summarizes normalized entropy of window loads. CV is sample standard deviation divided by mean load. Gini summarizes load inequality. This CV is distinct from technical-replicate quantification CV.

U is uncovered probability under the fitted precursor-count distribution. Files store a 0–1 fraction and the GUI displays a percentage. $Sw = Hnorm - 0.15 \times CV - 0.15 \times Gini - 0.25 \times U$. These model scores do not measure identification gains.

Each run supplies a fresh count-weighted fit. Packaged S3 geometry and quantile constraints then guide candidate generation. Canonical merging requires the recorded proximity, topology, logical-geometry and raw-domain checks.

Keep result.json, scheme_metrics.tsv, evaluation/fitted_model.json, fit-provider receipts, merge_decision.json and the final window table. Model diagnostics and any raw-IM projection remain part of interpretation.

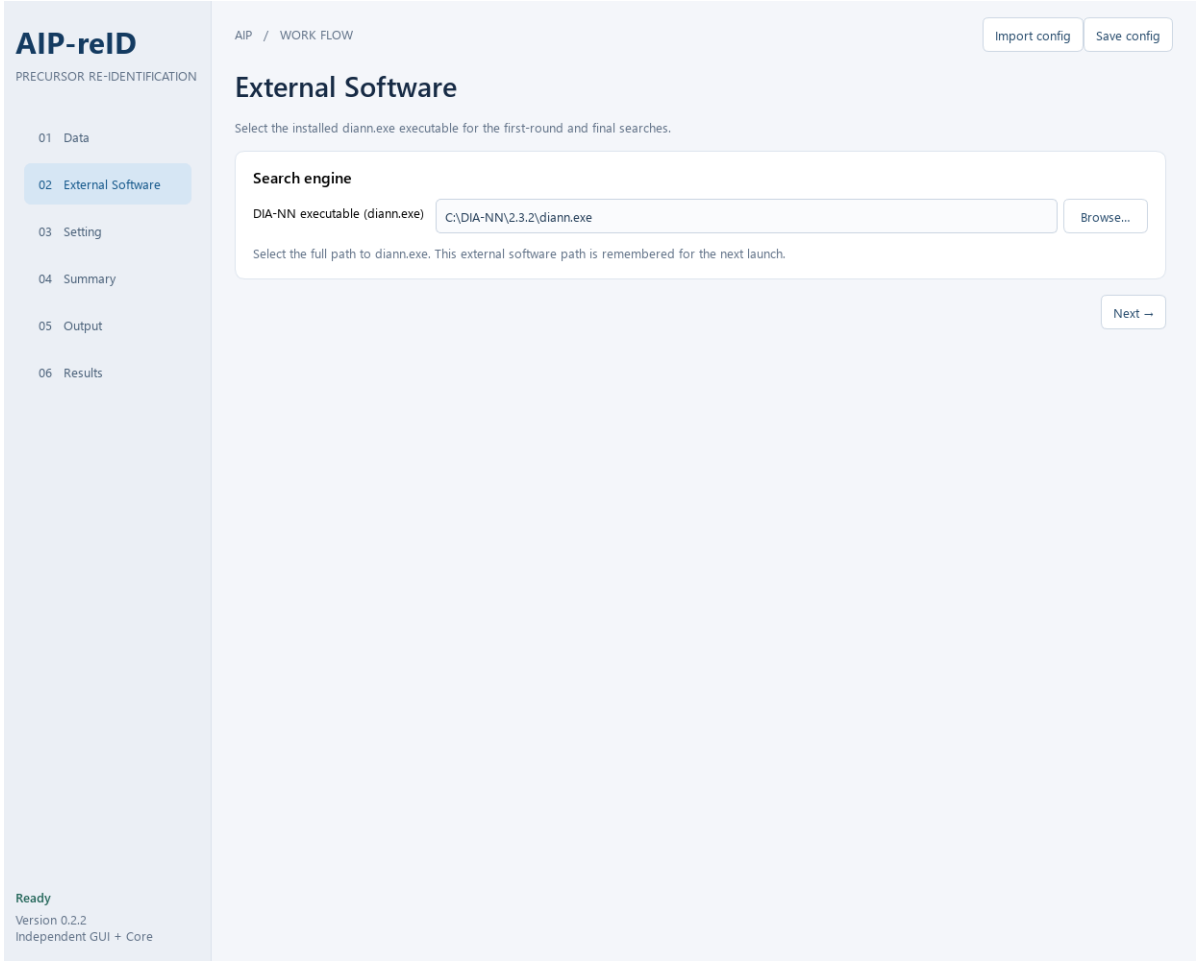
17. AIP-reID input data



1. Normal startup clears prior analysis inputs. Use Import config or Open_Demo.ps1 to fill them explicitly.
2. On Data, select the acquired .d folder and the FASTA database appropriate for the sample.
3. Click Next. Execution generates the complete predicted library from FASTA using the selected settings.

Actual Desktop0.2.2 inspection of the completed run1936 example. No new GUI analysis started. Only the External Software executable path is remembered between sessions.

18. AIP-reID external software



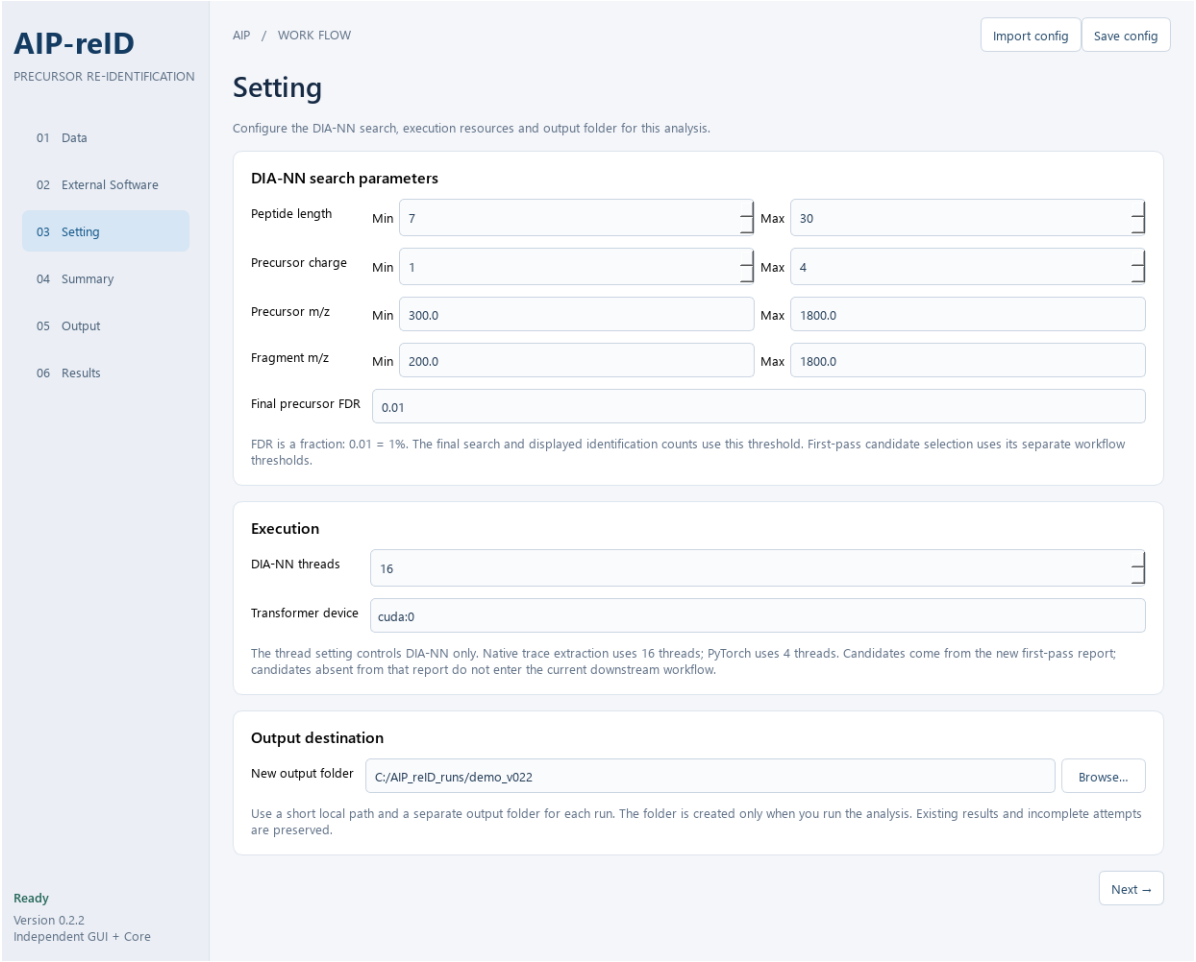
1. On External Software, select diann.exe from a complete DIA-NN 2.3.2 installation.

2. Keep its support files together. The Core resolves timsdata.dll and the contaminant FASTA beside DIA-NN.

3. The chosen executable path is remembered for this Windows user. Click Next to review analysis settings.

The external executable path is separate from analysis parameters. Importing a configuration can explicitly replace it. Check the path again after moving or reinstalling DIA-NN.

19. AIP-reID search and output settings



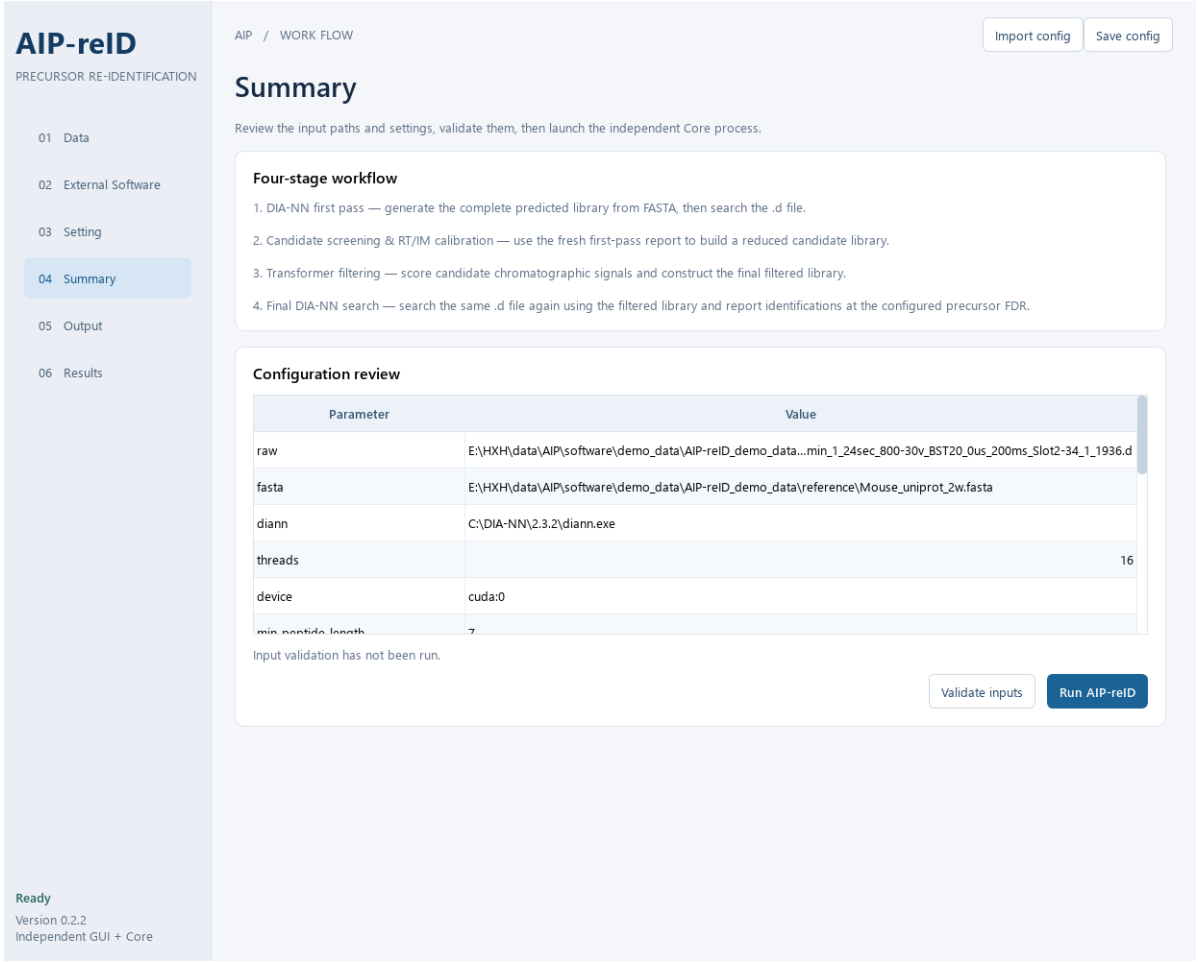
1. Set peptide length, charge, precursor m/z and fragment m/z ranges for DIA-NN.

2. Set Final precursor FDR, DIA-NN threads and Transformer device. FDR 0.01 means 1%.

3. Under New output folder, choose a fresh, short local path, then continue to Summary.

DIA-NN 2.3.2 defaults: length 7–30, charge 1–4, precursor m/z 300–1800, fragment m/z 200–1800, FDR 0.01. First-pass selection uses separate workflow thresholds.

20. AIP-reID validation and execution



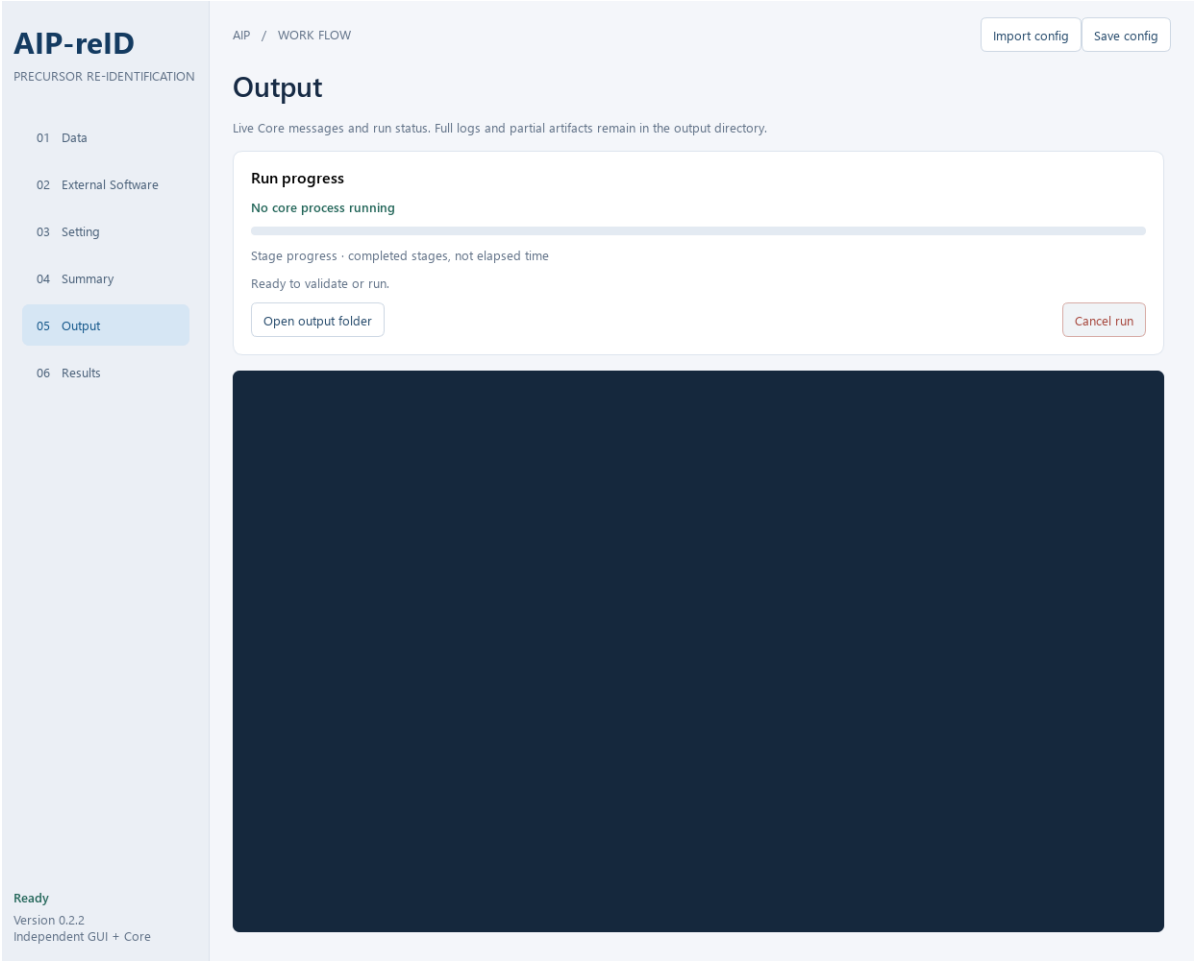
1. On Summary, review the input paths, search parameters and output folder chosen on Setting.

2. Click Validate inputs and resolve errors. If RAW or output paths change, repeat the helper's .quant path check.

3. Click Run AIP-reID. The workflow generates prediction, searches, filters candidates and searches again.

Open_Demo.ps1 normally uses C:/AIP_reID_runs. -OutputRoot selects a shorter root. Keep both .quant paths within 250 characters and UTF-8 bytes. Capture: inspection only.

21. AIP-reID progress and cancellation



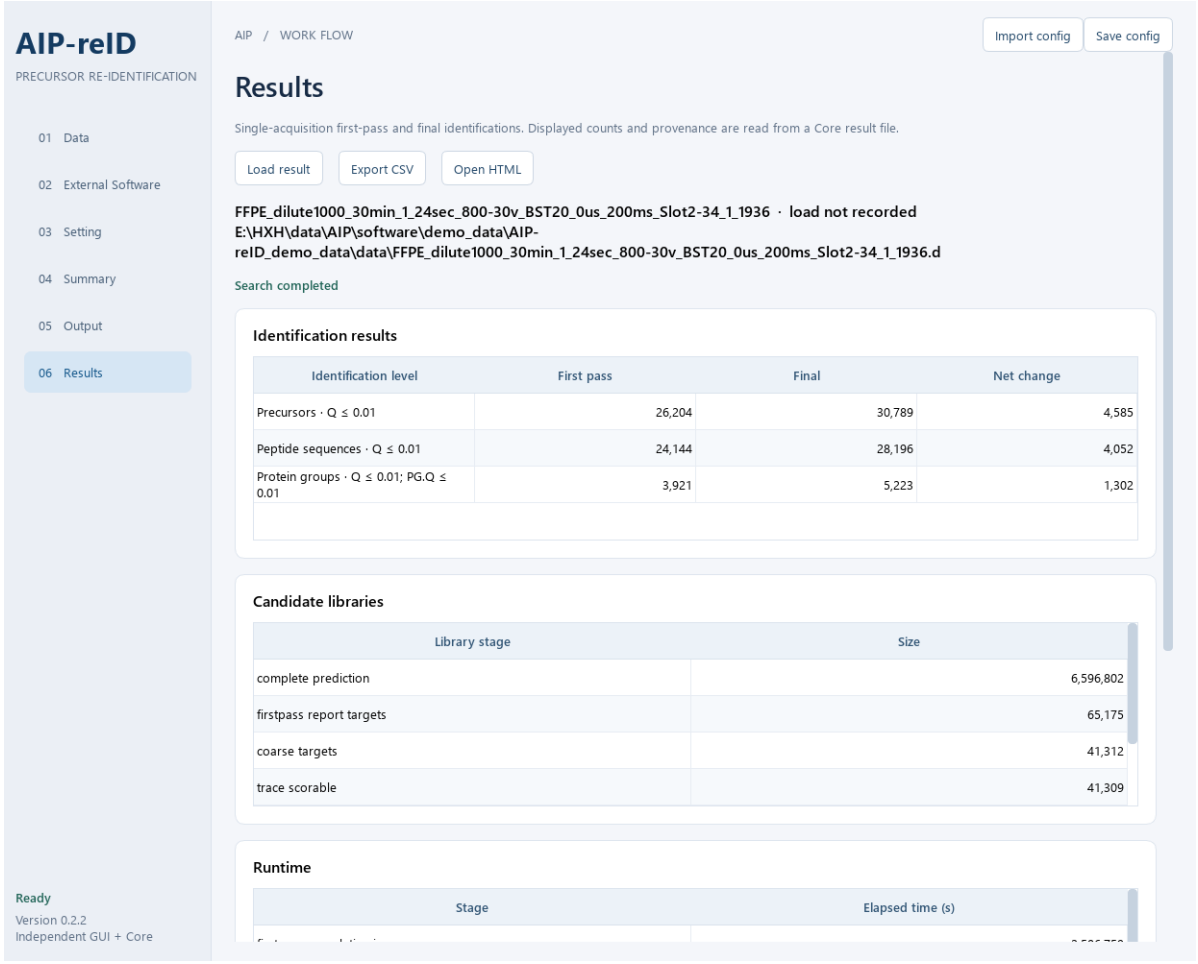
1. Open Output to follow stage messages and the live Core log during a new GUI run.

2. Use Open output folder to inspect detailed DIA-NN logs and intermediate artifacts.

3. Cancel run stops the Core process tree. Keep partial files. This Desktop workflow has no resume; retry in a fresh short folder.

Idle GUI after loading the completed example. It did not run the analysis or show the separate CLI job live. Progress measures completed stages, not elapsed time.

22. AIP-reID result display



1. A successful current search opens Results. Use Load result to inspect a saved Desktop result.json.

2. Check the run identity and recorded threshold before comparing First pass, Final and Net change.

3. Use Export CSV or Open HTML for summaries. Review Candidate libraries, Runtime and Output files.

Desktop0.2.2 loads the completed Core0.2.0 r2 example at Q.Value ≤ 0.01. Its recorded threshold stays with the result. New searches clear old counts.

23. AIP-reID counts and result interpretation

The historical run1936 r2 example used precursor Q.Value ≤ 0.01. Its final report contains 30,789 target precursors, 28,196 stripped peptides and 5,751 canonical protein-group strings.

With the additional PG.Q.Value ≤ 0.01 filter: 30,261 precursors, 27,668 stripped peptides and 5,223 protein groups. The GUI protein-group count uses this additional filter. New runs retain their selected final precursor FDR.

Library sizes: complete prediction 6,596,802; coarse candidates 41,312; Transformer-scored 41,309; final library 40,711. At target Q ≤ 0.01, final precursors increased by 4,585 from first pass. Counts come from each report separately.

Core0.2.0 produced this example. Desktop0.2.2 loads it without a new scientific run. Computational checks passed. Independent scientific validation remains outside this demo. No historical reports or ID lists fed the search.

24. Common errors and recovery

Module unavailable: keep the complete suite directory together. Use Path or Module settings if choosing an external installation. Check each module GUI, Core and runtime directory before retrying.

Input rejected: check raw format, required report columns and exact run identity. WES needs measured inverse mobility and valid raw calibration. reID needs FASTA and a complete DIA-NN installation.

WES review required: expand Model diagnostics and read the log. Numerically usable fits may yield reference-calibrated candidates despite failed scientific checks. The inactive legacy override does not bypass data or geometry errors.

reID .quant failure: choose a short local -OutputRoot with Open_Demo.ps1. Preserve failed files and confirm the old writer stopped. This Desktop workflow has no resume: retry the complete pipeline in a fresh directory.

25. Saved files and documented versions

AIP-Scope 1.4.0: root AIP-Scope.exe opens the three module folders, user guide and demo directory. %LOCALAPPDATA%/AIP-Scope-Launcher holds paths, launch.log, the encrypted model profile and saved Q&A sessions.

AIP-MSRefine 0.1.1: modules/AIP-MSRefine. Retain manifest.json, configuration.json, run.log and MSCohort_<run> with native tables, figures and the HTML report.

AIP-WES Desktop 0.5.6 / Core 0.5.1-ref2: modules/AIP-WES. Retain result.json, evaluation, fitted model, window tables, figures, fit-provider receipts and reference/merge/raw-domain diagnostics.

AIP-reID Desktop / Core 0.2.2: modules/AIP-reID. Keep the whole run directory and its recorded search settings. Demo inputs are under demo_data. The completed r2 example used Core0.2.0; new runs need a separate short output root.

Appendix A. MSRefine Spectronaut fields

Run and identification: R.FileName, EG.PrecursorId, EG.ModifiedPeptide, PEP.StrippedSequence, PG.ProteinGroups, EG.Qvalue, PG.Qvalue, EG.IsDecoy.

Quantification and precursor properties: PG.Quantity, PEP.Quantity, EG.TotalQuantity (Settings), EG.SignalToNoise, FG.Charge, FG.PrecMz.

Chromatography: EG.StartRT, EG.EndRT, EG.ApexRT, EG.FWHM, EG.PeakWidth, EG.DatapointsPerPeak, EG.DatapointsPerPeak (MS1).

Additional required fields: FG.RawMassAccuracy (PPM), FG.PrecWindow, FG.PrecWindowNumber, PEP.NrOfMissedCleavages. Export all listed fields to TSV/TXT and run Validate inputs before analysis.

Appendix B. MSRefine DIA-NN fields

Run identity: Run or File.Name. Precursor identity: Precursor.Id, Modified.Sequence, Stripped.Sequence, Precursor.Mz, Precursor.Charge.

FDR and quantification: Q.Value, PG.Q.Value, Precursor.Normalised, PG.MaxLFQ. Protein annotation: Protein.Group, Protein.Ids.

Retention time: RT, RT.Start, RT.Stop. Values use minutes. Export a tab-separated TSV/TXT report with all required columns.

Use FDR 0.01 for this parser. FWHM is derived as (RT.Stop − RT.Start)/2. Unsupported DIA-NN S/N and mass-error measurements remain unavailable in the structured result.

Appendix C. AIP-WES report fields

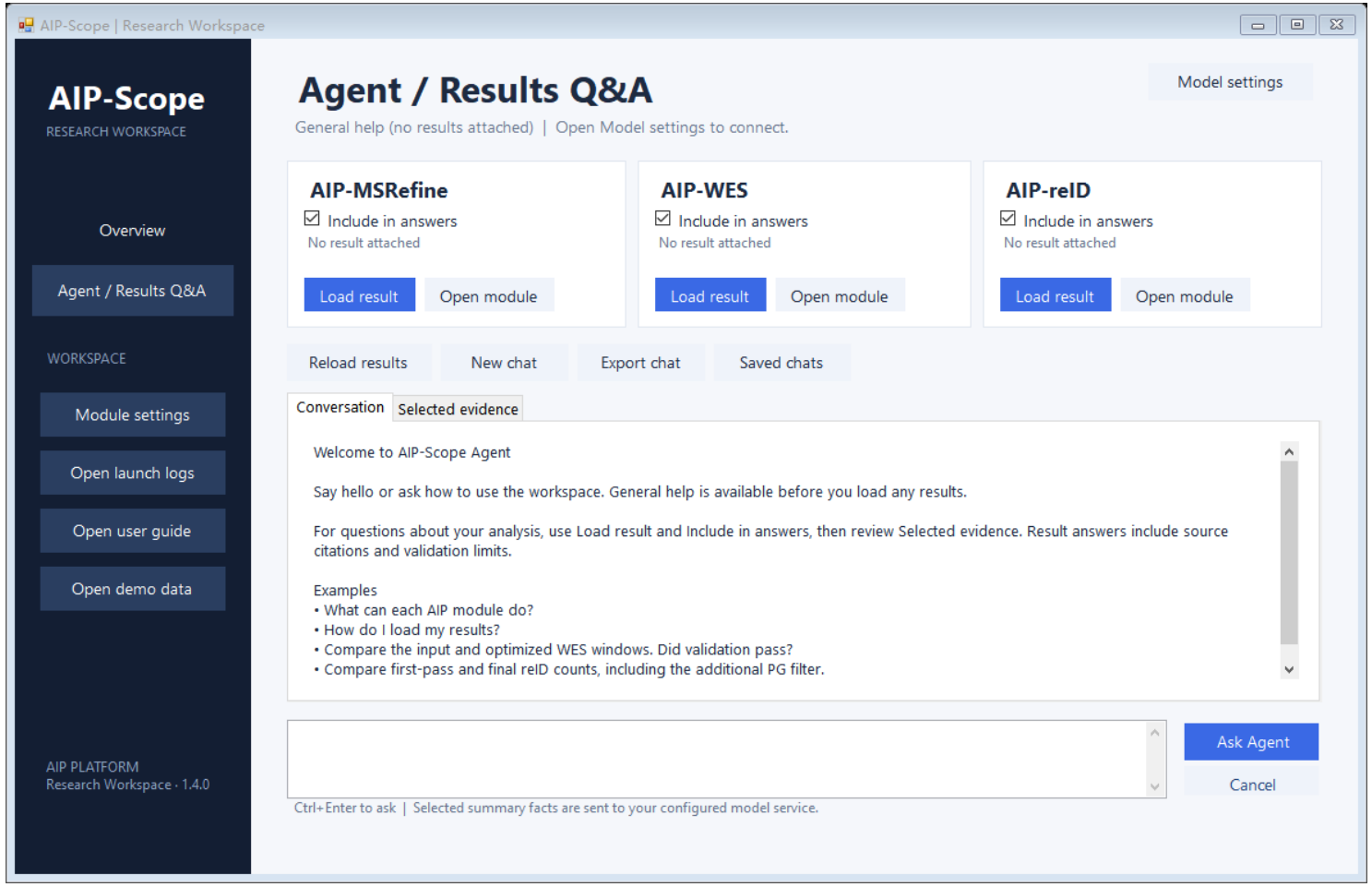
DIA-NN: Run, Modified.Sequence, Precursor.Charge, Q.Value and IM. Precursor.Id, Precursor.Mz, Decoy/Is.Decoy and RT can supply additional identity, mass and target-status information.

Spectronaut: R.FileName, EG.PrecursorId or modified peptide plus charge, EG.Qvalue and observed EG.IonMobility or FG.ApexIonMobility. FG.PrecMz is recommended.

Use measured inverse reduced mobility rather than predicted IM, iIM or iRT. Supported report containers include TSV, CSV and Parquet. Check the detected format and reported field mapping.

The fitter uses unique charge 2+ and 3+ precursors at $q \leq 0.01$ with unit count weights. Each charge class needs adequate observations. If a decoy column is absent, verify that the export contains targets only.

26. Results Q&A workspace



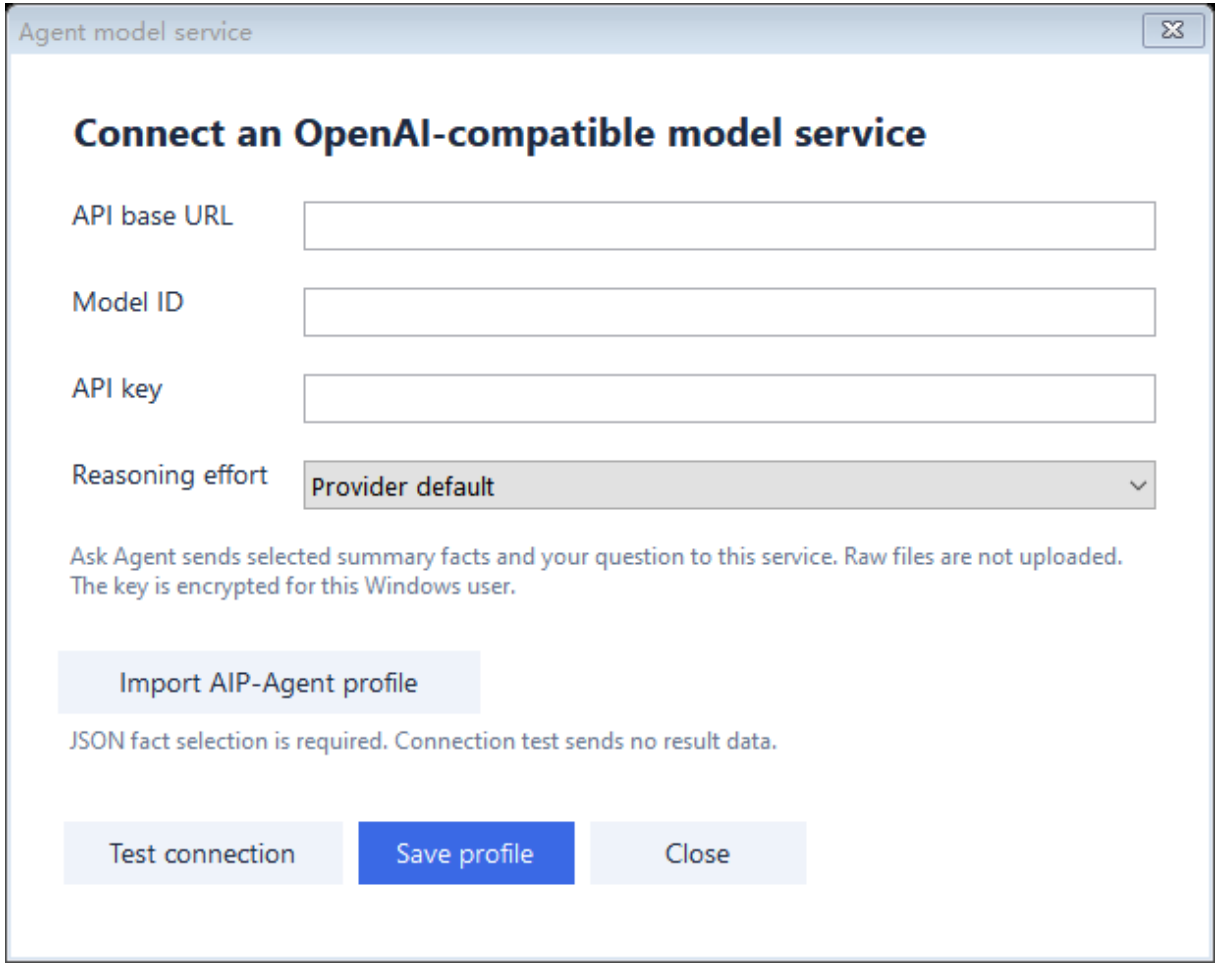
1. Open Agent / Results Q&A. With no included results, a configured model can provide general workspace help.

2. Load module summaries and choose Include in answers when asking about measurements or a particular completed analysis.

3. Review Selected evidence, enter the question and select Ask Agent. Open module returns to full plots and analysis controls.

General help carries no result evidence. A question about measured counts or scores requires the corresponding module summary.

27. Model service setup



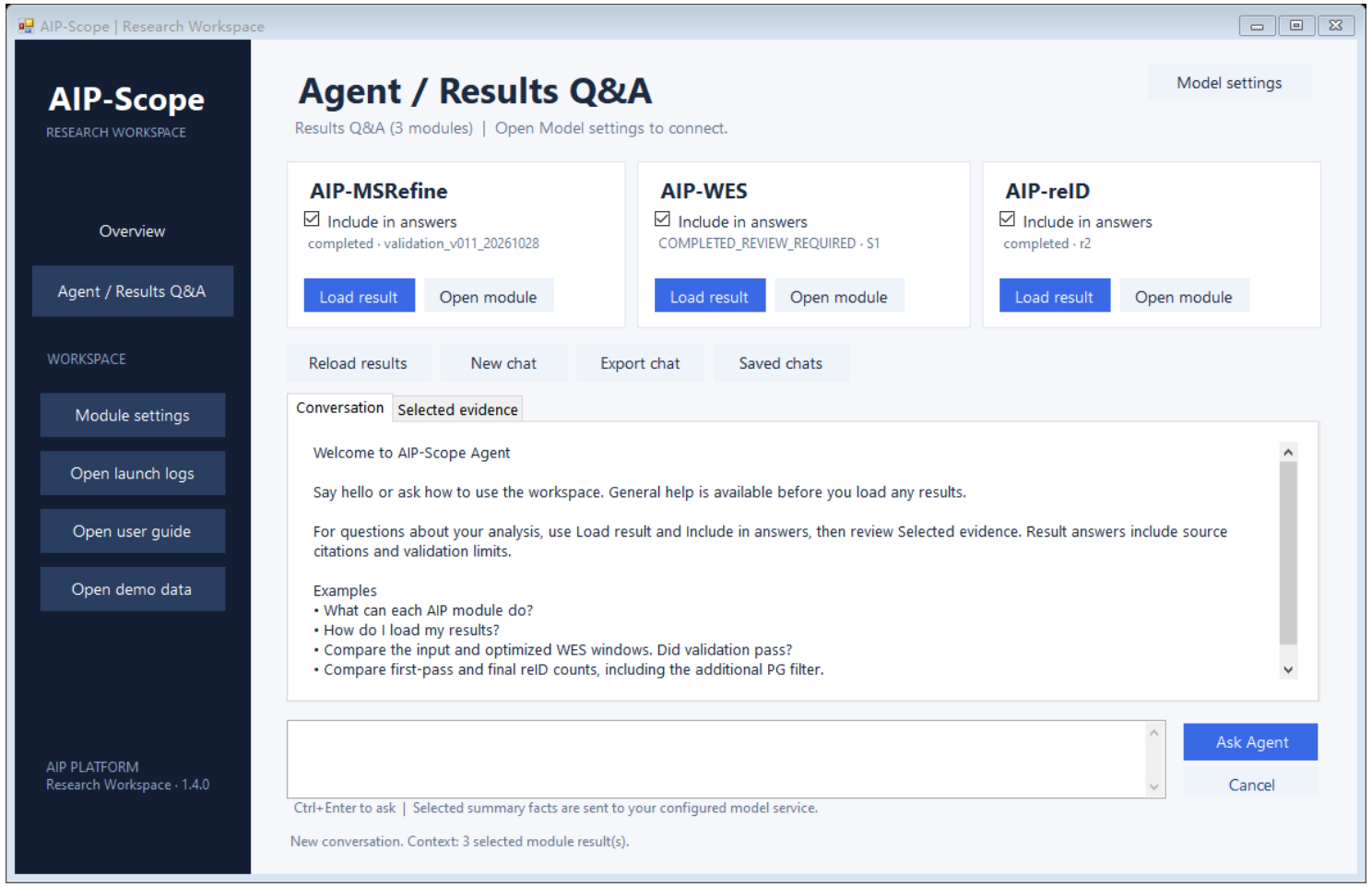
1. Open Model settings. Enter the API base URL, model ID and API key for a compatible service.

2. Alternatively, use Import AIP-Agent profile and select its var/platform.sqlite file.

3. Click Test connection, then Save profile. Leave reasoning effort at Provider default unless supported.

The key is encrypted for this Windows user. Test connection sends no result data. This capture shows an unconfigured dialog.

28. Attach module results



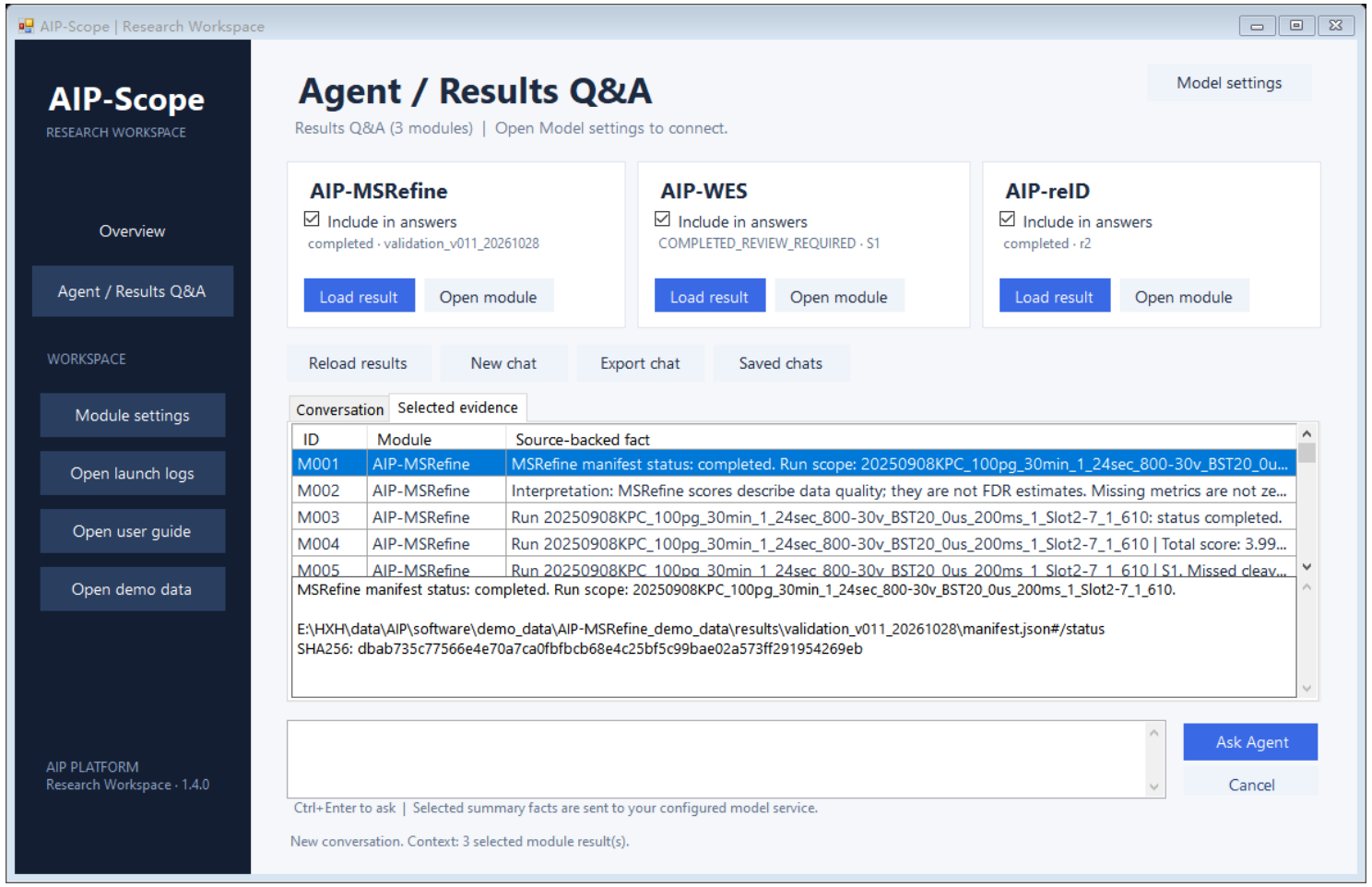
1 For AIP-MSRefine, click Load result and select the completed output's manifest.json.

2 For AIP-WES or AIP-reID, load the module's result.json. Use the normalized reID Desktop summary.

3 Check Include in answers for the required modules. Confirm each run and read its recorded status.

Actual 1.4.0: MSRefine run610, WES S1/run2333 and reID run1936 r2. These are separate acquisitions; completed execution does not clear scientific warnings.

29. Review selected evidence



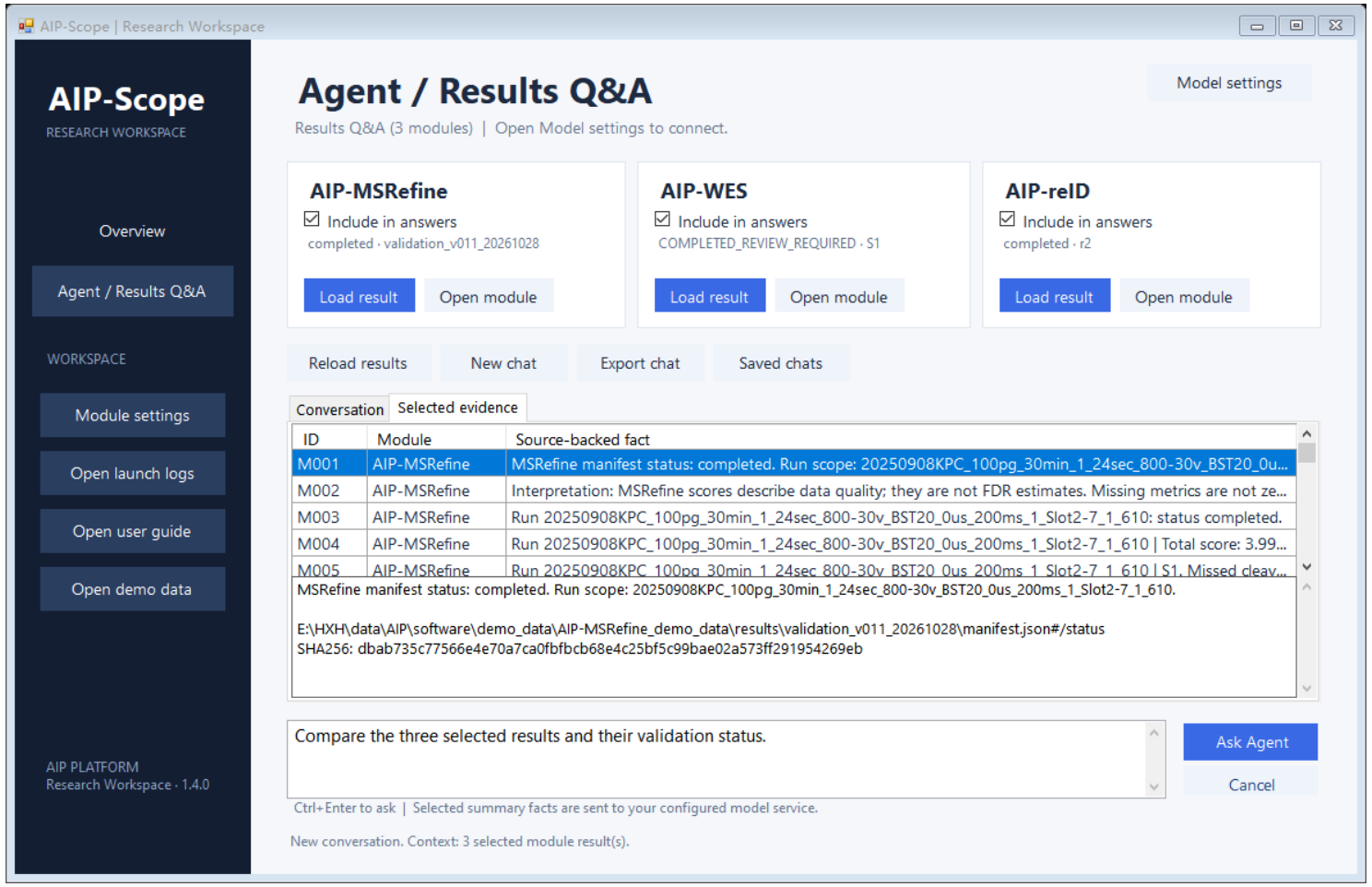
1. Open Selected evidence to review the facts available from the included modules.

2. Select a row to inspect its full text, source file, JSON field location and SHA256 hash.

3. Confirm the run identity, metric units and validation state. Exclude unrelated module results.

Evidence IDs identify facts in the current selection. Missing measurements remain unavailable; results from different runs remain distinct.

30. Ask and review answers



1. Include the relevant module summaries, then enter a question about their recorded facts.

2. Review Selected evidence before choosing Ask Agent or Ctrl+Enter. A configured service is needed to submit the question.

3. When an answer arrives, check its evidence IDs, Context and limitations, and Sources. Cancel stops a pending request.

Actual AIP-Scope 1.4.0 with a question entered but not submitted. No model response, external request or scientific execution is represented by this capture.

31. Questions for each module

AIP-MSRefine: "What does the total score mean? Which metrics are unavailable?" Ask about a specific metric to retrieve its value, unit, recorded status and definition.

AIP-WES: "Compare current and candidate windows and Sw. Did fit validation pass, and was the packaged reference applied?" Keep model scores and reference matching separate from measured acquisition benefits.

AIP-reID: "Which precursor threshold produced these counts? What changes after the additional PG filter?" Confirm the run identity, recorded thresholds and counts in the actual search reports.

Across modules: "Which limitations remain in these selected results?" Confirm sample and run identities. English or Chinese questions are accepted. Source-backed rendered answers and the interface are English.

32. Follow-up, reload and export

Ask a follow-up in the same conversation. The previous three user questions help clarify its context. The current selected facts remain the source of the answer. New chat clears the visible conversation.

After a module updates its output, select Reload results. The Agent checks source hashes before sending a question and before publishing an answer. Loading, reloading or changing included modules starts a new conversation.

Use Export chat to save a readable TXT transcript or JSON source receipts. JSON records the question, answer, model, cited fact IDs, source paths and hashes for review.

Successful answers are also saved locally in %LOCALAPPDATA%/AIP-Scope-Launcher/sessions. Saved chats opens this folder. These files preserve past answers; they are not automatically restored as context for a new question.

33. Answer limits and troubleshooting

Model unavailable: check the API URL, model ID and key in Model settings, then test the connection. Authentication needs valid credentials; rate limits require a later retry. Module analysis does not require Agent access.

Missing or changed result: load the correct summary and reload it after updates. For reID, choose Desktop result.json rather than scientific_result.json. The recorded incomplete status and unavailable values remain visible.

Without results, the Agent provides labeled general help. Evidence-based questions require selected summary facts. It does not inspect figure pixels, start scientific analyses or invent unsupported measurements and causes.

Selected summary facts and recent questions are sent to your configured model service. RAW files and figures are not uploaded. Successful execution, reference matching or a higher score does not establish scientific validation.